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The Recoverability Frontier of Damaged Information-Bearing Systems

Coding, topology, reachability, linear estimation and thermodynamic cost in one stylised framework

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Scope. This paper studies abstract, dimensionless models of damaged information-bearing systems. It asks when the information in a damaged system can still be identified, decoded, reached by a control action, estimated from partial observations, and restored at a finite physical cost. **It does not study the reversal of death, the revival of any organism or person, personal identity, or any medical intervention.** Every number is produced by `code/recoverability_experiments.py` with the fixed seed 20261108. No experiment on living material is reported, and no claim about biology follows from these results without a separate measurement programme.

Abstract. When a system is damaged, three questions are often run together: is the information still present, can a controller drive the system back to a functional state, and what does that cost? We separate these questions and give each a precise meaning. Information recovery is modelled as decoding of a redundant erasure code. Identifiability of organisation is modelled through persistent clusters of a point cloud. Functional restoration is modelled as reachability and minimum-energy control of a linear system. Partial observation is modelled as Gaussian linear estimation, and the physical floor on restoration as a Landauer-type bound. We prove six propositions, including an exact closed form for full-rank recovery of random binary codes, a generic reachability rank of min(n, Tm), and monotonicity of recoverability under damage. The numerical results show four things. A coding threshold sits at $1 - R$: the measured 50% points are 0.73, 0.48 and 0.23 against 0.75, 0.50 and 0.25. A topological signature survives deletion of 87% of points but is fragile to measurement jitter at a fixed scale. Reachability has a cliff at $m = \lceil n/T \rceil$ actuators, and the energy cost just above the cliff is about three orders of magnitude above full actuation. The observed fraction needed for estimation depends strongly on correlation, falling from 90% at $\rho = 0.2$ to 40% at $\rho = 0.9$. A composite frontier shows that the control layer acts as a cliff rather than a slope. We state a falsifiable hypothesis for each layer, report which survived and which failed, and specify what is needed to test the composite conjecture end to end. The contribution is a framework, a set of verified sub-results, and two negative findings that constrain how the framework should be used.

Keywords: erasure codes · persistent homology · reachability · minimum-energy control · linear estimation · rate–distortion · Landauer bound · recoverability

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1 Introduction and scope

Damage to a biological or engineered system is usually described as a loss of material, a loss of function, or both. Those descriptions answer a different question from the one that governs recovery. A system can lose most of its physical parts and still hold the information needed to rebuild them, provided the surviving parts are arranged so that reconstruction is possible. The same system can keep most of its parts and still be unrecoverable, because the surviving information has become inaccessible, uncontrollable or too costly to use. Recovery therefore depends on at least four logically distinct properties: whether the information survives, whether it can be estimated from what remains, whether a controller can reach a functional state, and what physical resources that requires. A fifth quantity, the physical floor on restoration, bounds all four.

This paper formalises these five quantities as layers of one framework and tests the layers numerically on stylised systems. The central object is the set of conditions under which all layers hold together, which we call the recoverability frontier. We treat the frontier as a mathematical and computational question, not as a fact about any organism.

Scope and exclusions. We do not model the death of any organism, the restoration of any deceased animal or person, the preservation or revival of personal identity, or any clinical procedure. *Recoverability* here refers only to the recovery of an abstract information state in a simulated system. Connecting these results to biology needs a separate measurement programme, outlined in Section 13.

How to read the evidence. Statements are labelled Proposition (proved in Section 4 or Appendix A), Conjecture (not proved), Numerical result (produced by the code with the stated seed and sizes), or Hypothesis (a stated prediction with a status). Where an outcome contradicts what we expected, we say so.

2 Motivation, and what existing evidence does and does not show

Four bodies of work motivate the layers. Information theory gives the limit of what a redundant code can recover (Shannon 1948; Cover and Thomas 2006), and the theory of sparse-graph codes shows how decoding succeeds or fails sharply as damage grows (Richardson and Urbanke 2008). Persistent homology describes which multiscale structures survive perturbation (Edelsbrunner, Letscher and Zomorodian 2002; Zomorodian and Carlsson 2005; Carlsson 2009). Control theory gives exact conditions under which a system can be steered (Kalman 1963; Sontag 1998). Thermodynamics gives the cost of information processing (Landauer 1961; Bennett 1982; Seifert 2012). Each is standard. Our contribution is to place them in one framework with a common notion of damage, and to test their interaction.

Some experimental work bears on the question without answering it. Reports of partial restoration of circulation and cellular function in post-mortem large-animal tissue (Vrselja et al. 2019) show that certain processes can be restarted after an interval. Those reports do not show restoration of organised, conscious or whole-body function, and we do not treat them as evidence for organismal recovery. Memory research shows that engram representations are distributed and that the physical substrate of memory is still under study (Josselyn and Tonegawa 2020). These findings motivate the question of which information survives damage. They do not supply thresholds, and we do not use them to set parameters.

3 Formal framework: five quantities of recoverability

Let X_0 denote the intact system, a random object whose state includes a redundant encoding of target information $Z = \Phi(X_0)$. A damage process of severity $x \in [0, 1]$ returns a damaged state X_x. An observation channel returns $Y = H(X_x) + \varepsilon$, where H is a linear or sampling map and ε is noise. The five quantities are:

Definition 1 (information recoverability). $D^*(x) = \inf$ over estimators $\hat{Z}(Y)$ of $E[d(Z, \hat{Z})]$, the minimum achievable distortion. Information is recoverable at severity x when $D^*(x) \le D_{max}$.

Definition 2 (code recoverability). When redundancy is an erasure code of rate R and damage erases a fraction x of coded symbols, $P_{code}(x; R)$ is the probability that all information symbols are decoded from the survivors.

Definition 3 (topological recoverability). When organisation is encoded by single-linkage clustering of a point cloud, $P_{topo}(x)$ is the probability that the number of clusters at a fixed scale is unchanged by damage of severity x.

Definition 4 (dynamical recoverability). Let the damaged system evolve as $x_{t+1} = A\,x_t + B_a\,u_t$, where B_a retains the surviving actuators after a fraction a is lost. The target is reachable in horizon T if the controllability matrix has rank n. $P_{ctrl}(a)$ is the fraction of actuator sets for which the target is reachable and the minimum input energy stays within κ times a reference.

Definition 5 (thermodynamic floor). Restoring b bits at temperature T costs at least $W_{min} = k_B\,T \ln 2 \cdot b$ (Proposition 6).

The recoverability frontier. Under an independence assumption, the composite success probability is $P_{rec}(x, a; R) = P_{code}(x; R) \cdot P_{topo}(x) \cdot P_{ctrl}(a)$, and the frontier is the set of parameters where this is at least one half. Section 13 explains why the product is an approximation and not a derived result.

$$\Phi = \{\ (x, a, R) : P_{rec}(x, a; R) \ge 1/2\ \}$$

Hypothesis H0 (composite threshold). The layers jointly determine recoverability, and the frontier Φ separates three regimes: information survives and function is restorable; information survives but no admissible control reaches the target; and observations no longer identify the target. We test the layers separately and test the composite only under the stated independence assumption.

4 Propositions

Full proofs of Propositions 2 and 3 are in Appendix A. Propositions 1, 4, 5 and 6 are short and proved in place.

Proposition 1 (monotonicity under damage). If X_x is obtained from X_0 by a damage channel that does not depend on Z, then $I(Z; Y) \le I(Z; X_x) \le I(Z; X_0)$ for any observation $Y = H(X_x) + \varepsilon$. Consequently $D^*(x)$ is non-decreasing in x.

Proof. The data-processing inequality states $I(Z; g(W)) \le I(Z; W)$ for any measurable g when $Z \to W \to g(W)$ is a Markov chain (Cover and Thomas 2006, Thm. 2.8.1). Because the damage does not depend on Z, the chain $Z \to X_0 \to X_x \to Y$ is Markov, which gives both inequalities. A larger mutual information allows a smaller minimum distortion, so D^* is non-decreasing in x. □

Proposition 2 (exact full-rank probability of random binary codes). For a uniformly random $k \times n$ binary matrix G and a uniformly random set S of s columns, $P(\text{rank } G_S = k)$ equals

$$P_{full}(k, s) = \prod_{i=0}^{k-1} (1 - 2^{i-s}) \text{ for } s \ge k, \text{ and } 0 \text{ otherwise.}$$

Hence $P_{full} \to 1$ when $s - k \to +\infty$ and $\to 0$ when $s - k \to -\infty$, so the erasure threshold of random linear codes of rate R is $1 - R$ in the large-n limit, the Shannon limit for the binary erasure channel (Shannon 1948).

Proposition 3 (generic reachability rank). For $x_{t+1} = A\,x_t + B\,u_t$ with n states and m inputs, the horizon-T controllability matrix $M = [A^{T-1}B, \ldots, B]$ has rank at most $\min(n, Tm)$. For generic (A, B) the rank equals $\min(n, Tm)$. The target is therefore generically reachable in horizon T if and only if $Tm \ge n$, and the generic cliff is at $m^* = \lceil n/T \rceil$.

Proposition 4 (minimum-energy steering). If M has full row rank n, the minimum input energy $\Sigma_t \|u_t\|^2$ that steers x_0 to z in horizon T is

$$E_{min} = r^\top (M M^\top)^{-1} r,\ r = z - A^T x_0.$$

Proof. The terminal state is $x_T = A^T x_0 + M u$, with u the stacked inputs. The minimum-norm solution of M u = r is $u^* = M^\top (M M^\top)^{-1} r$, with $\|u^*\|^2 = r^\top (M M^\top)^{-1} r$. Any other solution differs from u* by a component in the null space of M, which is orthogonal to u* and so only adds norm. □

Proposition 5 (more observations cannot hurt linear recovery). For a Gaussian vector Z with covariance Σ and noisy observations on index sets $S \subseteq S'$, the LMMSE error $\mathrm{tr}(\Sigma - K_{S'})$ is no larger than $\mathrm{tr}(\Sigma - K_S)$, where $K_S = \Sigma_{\cdot S}(\Sigma_{SS} + \sigma^2 I)^{-1} \Sigma_{S\cdot}$.

Proof. K_S is the covariance explained by observing S. Under Gaussian conditioning, explained covariance grows in the Loewner order as observations are added, so the residual trace shrinks. □

Proposition 6 (Landauer-type floor, stated as a bound). Suppose a restoration process raises the information held about Z by b bits at isothermal temperature T, with the entropy reduction paid for by an environment. Then the work W satisfies $W \geq k_B T \ln 2 \cdot b$. This is the standard Landauer argument extended to information gain through the second law (Landauer 1961; Bennett 1982; Sagawa and Ueda 2008). It is a bound under explicit assumptions, and we do not claim that real processes operate near it.

5 Layer A: coding recoverability

We take n = 48 coded symbols and rates $R \in \{0.25, 0.5, 0.75\}$, giving k = 12, 24 and 36 information symbols. For each erasure fraction ε from 0 to 0.9 in steps of 0.05 we draw 400 independent random binary codes, erase the stated fraction of symbols, and test whether the information is recovered by GF(2) elimination.

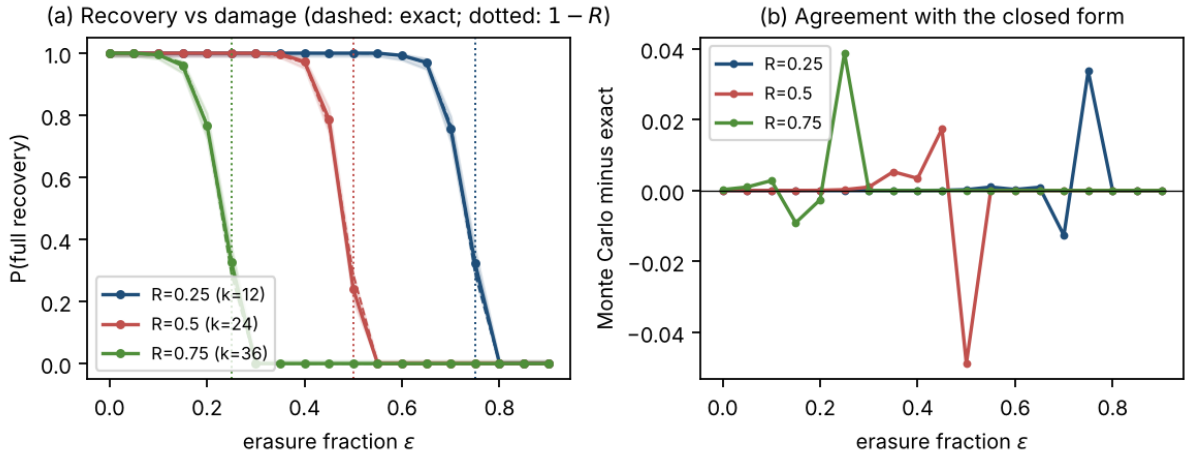


Figure 2. Full-recovery probability of random binary linear codes. (a) Monte Carlo estimates with 95% Wilson bands (points), exact closed form of Proposition 2 (dashed), and the Shannon limit 1 − R (dotted). (b) Monte Carlo minus exact. Deviations are of the order of the sampling standard error at 400 trials, at most 0.025.

Rate R	Info. symbols k	Measured 50% point	Shannon limit 1 − R	Max |MC − exact|
0.25	12	0.730	0.75	0.034
0.5	24	0.476	0.50	0.049
0.75	36	0.230	0.25	0.039

Numerical result. The measured 50% point lies within 0.03 of the Shannon limit at every rate. The Monte Carlo estimates agree with Proposition 2 across all 57 (rate, erasure) points, with the largest deviation 0.049, which is about two standard errors and within what sampling over 57 points produces. Redundancy is not free: the more of a unit's capacity goes to function, the smaller the damage it can survive. This is the trade-off the framework predicts, and here it is checked against an exact formula rather than an approximation.

6 Layer B: topological recoverability

We sample N = 240 points from three Gaussian clusters, 80 points each, standard deviation 0.5, at the vertices of a triangle with sides about 4. Organisation is encoded by single-linkage clustering. At cut height 1.0 the intact data form three clusters of at least five points each. The largest within-cluster link is about 0.65 and the smallest between-cluster link about 1.23, so the cut sits in a clean gap. (A first attempt at cut 1.5 merged two clusters, which is why the cut was lowered; the choice was made from the intact data alone.) Damage removes a fraction d of points, optionally followed by Gaussian jitter of standard deviation 0.3 on the survivors. There are 300 trials per damage level.

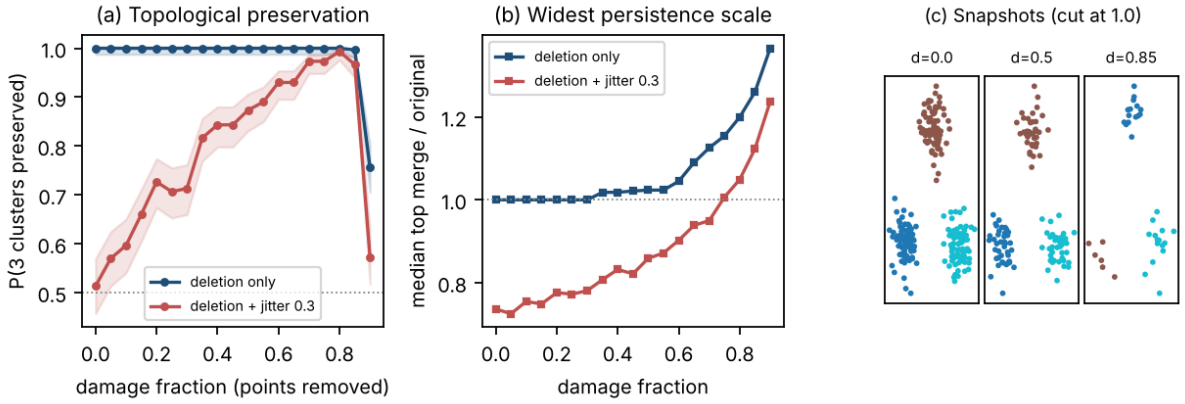


Figure 3. Topological recoverability of a three-cluster point cloud. (a) Probability that three clusters of size at least five survive, with 95% bands. (b) Median largest merge height relative to the intact value. (c) Snapshots at cut 1.0 after 0%, 50% and 85% deletion; colours are cluster labels.

Numerical result. With deletion alone, the three-cluster signature survives with probability above 0.99 until 85% of points are removed, and falls to 0.76 at 90% deletion. The 90%-preservation point is at d = 0.87. The largest persistence scale grows as points disappear, because clusters that lose members spread out, so the largest merge height reaches 1.37 times its intact value at 90% deletion.

Negative finding. Jitter of 0.3 on the survivors breaks the signature even when nothing is deleted: the preservation probability is 0.51 at d = 0. The cut of 1.0 was chosen for clean data, and jitter fills the gap between clusters. As deletion increases, the jitter curve rises, because thinner clusters are better separated relative to the noise. The practical lesson is that a topological invariant is meaningful only relative to a scale calibrated to the measurement noise of the particular system. Any comparison of a damaged state with an intact one must recalibrate the cut from data first.

7 Layer C: control reachability and energy

We build a random stable linear system with n = 20 states and spectral radius 0.9, with ten candidate actuators and horizon T = 8. For each number of active actuators m from 1 to 10 we sample 30 actuator sets and, for every reachable set, 10 random pairs of initial and target states. A set is reachable when the controllability matrix has rank n. We record the minimum input energy of Proposition 4, normalised by the median energy with all ten actuators. A set succeeds when it is reachable and its mean energy is at most twice that reference.

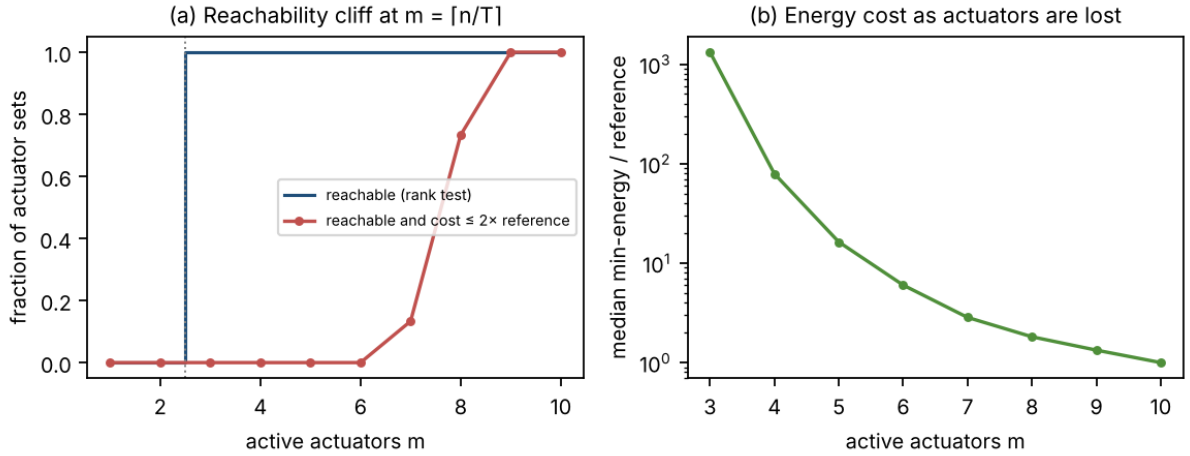


Figure 4. Control reachability as actuators are lost. (a) Fraction of actuator sets that are reachable (step) and fraction that succeed (points). The dotted line marks the generic cliff m* = ⌈n/T⌉ = 3. (b) Median minimum energy relative to full actuation, log scale.

Active actuators m	Actuator loss a	Reachable fraction	Success fraction	Median energy / reference
1	0.9	0.00	0.00	n/a
2	0.8	0.00	0.00	n/a
3	0.7	1.00	0.00	1317.52
4	0.6	1.00	0.00	79.00
5	0.5	1.00	0.00	16.31
6	0.4	1.00	0.00	6.01
7	0.3	1.00	0.13	2.84
8	0.2	1.00	0.73	1.82
9	0.1	1.00	1.00	1.33
10	0.0	1.00	1.00	1.00

Numerical result. The reachability rank matches Proposition 3. No set with m ≤ 2 is reachable and every set with m ≥ 3 is, so the cliff sits at m* = ⌈20/8⌉ = 3. Reachability is therefore a discrete property in this system, while energy is continuous and steep near the cliff: at m = 3 the median energy is 1318 times the full-actuation reference. Success is exceeded half the time only with at least eight actuators remaining. The consequence for practice is that a system can be controllable in principle and still be unrecoverable in practice, because the energy budget is exceeded well before the rank condition fails.

8 Layer D: information from partial observation

The damaged state is modelled as a stationary Gaussian vector Z in $\mathbb{R}^{40}$ with unit variance and correlation $\Sigma_{ij} = \rho^{|i-j|}$. It is observed on a random subset of coordinates with noise variance 0.1. The normalised LMMSE error D(o, ρ) is averaged over 40 random subsets for each observed fraction o. Proposition 5 implies that adding observations cannot increase D. We report the observed fraction needed for D ≤ 0.2 across correlation strengths from 0 to 0.98.

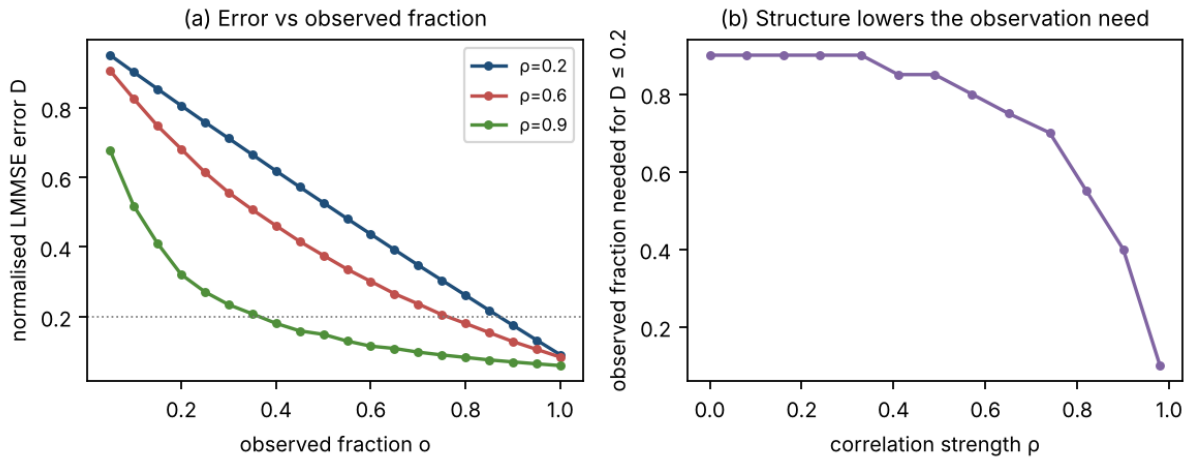


Figure 5. Information recoverability from partial observation. (a) Normalised LMMSE error against observed fraction for three correlation strengths; the dotted line marks D = 0.2. (b) Observed fraction needed to reach D ≤ 0.2, as a function of correlation strength.

Correlation ρ	Observed fraction for D ≤ 0.2	Interpretation
0.2	90%	weakly structured: most coordinates needed
0.6	80%	moderately structured
0.9	40%	strongly structured: under half needed

Numerical result. Structure lowers the observation requirement. A strongly correlated system can be estimated accurately from less than half its coordinates, whereas a weakly correlated one needs about 90%. At ρ = 0.98 the requirement falls to 10%, as Figure 5(b) shows. This supports the view that recoverability depends on joint structure, not on the amount of damage alone. The result holds for Gaussian linear estimation with known covariance, a strong assumption discussed in Section 13.

9 Layer E: thermodynamic floor

For a unit-variance Gaussian coordinate, reducing the normalised error to D requires at least ½ $\log_2(1/D)$ bits per coordinate (Gaussian rate–distortion; Berger 1971). Multiplying by k_B T ln 2 at T = 310 K gives a Landauer-type energy floor (Proposition 6). This is a floor, not a prediction of cost, and it does not depend on the mechanism that supplies the bits.

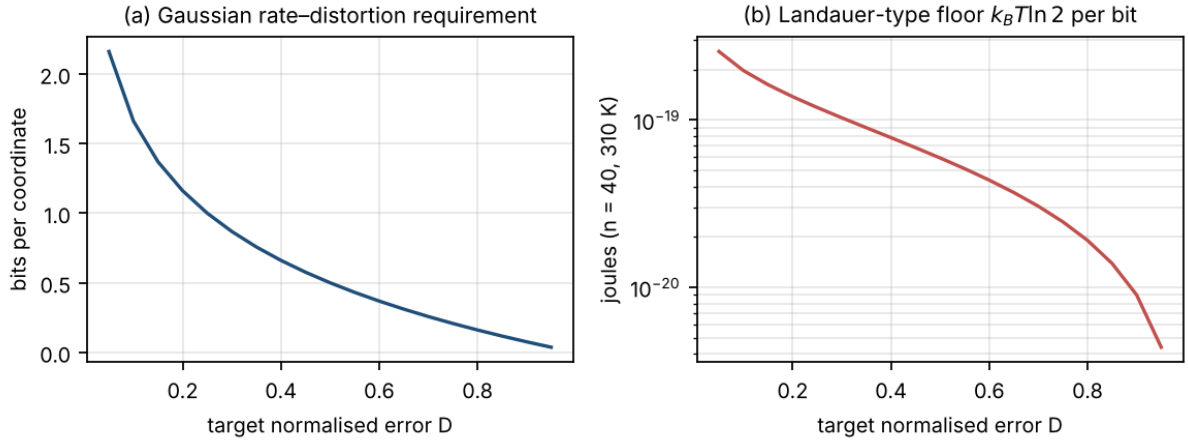


Figure 6. Thermodynamic floor for restoring information. (a) Bits per coordinate needed for a target normalised error D. (b) Total floor for n = 40 coordinates at 310 K in joules, log scale. The floor is a lower bound; no mechanism is assumed.

Numerical result. For D = 0.2 the requirement is 1.161 bits per coordinate. At 310 K this corresponds to k_BT ln 2 = 2.97e-21 J, so about 1.4e-19 J for 40 coordinates. For scale, the free energy of one ATP hydrolysis is roughly 8×10^{-20} J, about thirty times this floor per bit. The floor

therefore does not constrain restoration at these scales. The binding constraints in this framework are the reachability and energy conditions of Layer C. Real physical costs are set by dissipation far above the floor (Seifert 2012; Parrondo, Horowitz and Sagawa 2015).

10 The composite recoverability frontier

We combine Layers A, B and C under the independence assumption of Section 3: $P_{rec}(x, a; R) = P_{code}(x; R) \cdot P_{topo}(x) \cdot P_{ctrl}(a)$. Damage x is shared by the code and the point cloud, and actuator loss a enters through the control layer. Figure 7 shows the composite surface at R = 0.5 and the critical damage x*(a) for three code rates, where x*(a) is the largest damage with composite probability at least one half.

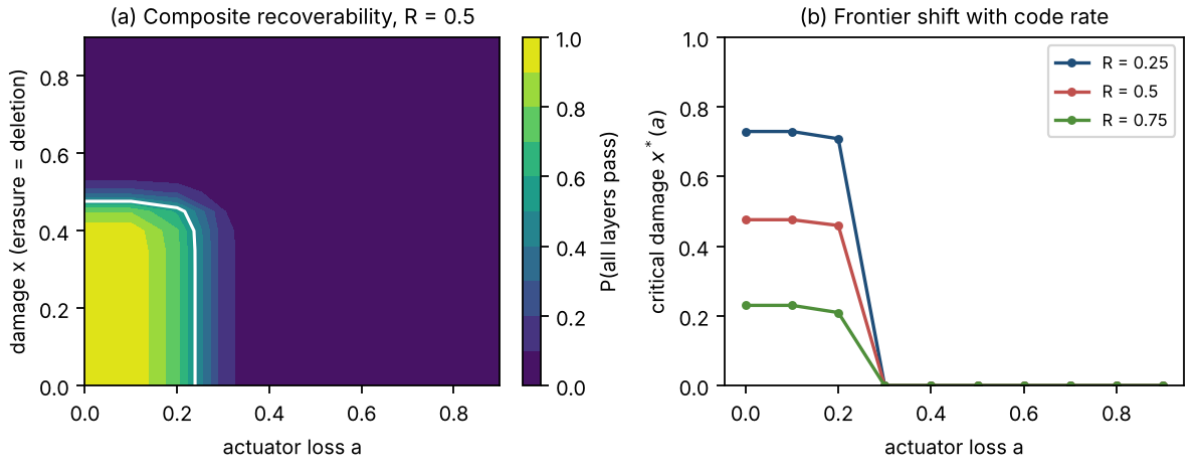


Figure 7. The composite recoverability frontier under the independence assumption. (a) Composite probability over damage x and actuator loss a at R = 0.5; the white contour is the one-half level. (b) Critical damage x*(a) for R = 0.25, 0.5 and 0.75. A value of 0 means no damage level is safe at that actuator loss.

Numerical result. At full actuation the critical damage is 0.73, 0.48 and 0.23 for R = 0.25, 0.5 and 0.75, roughly tracking 1 − R. The composite is set by the weakest layer. Topology is not limiting at these damage levels; control is. From an actuator loss of 0.3 onward no damage level is safe, because success falls below one half at all damage levels. The frontier position is therefore governed by a discrete event, and that is the main structural finding of the composite analysis.

This result must be read with the caveat of Section 13. The three layers were evaluated in separate simulations, and the product is correct only if the events are independent, which nothing guarantees. A joint simulation in which one damage process acts on the code, the point cloud and the actuators together is needed before the frontier can be treated as more than an illustrative composition.

Layer	Quantity	Critical value	Source
A Coding	erasure fraction at 50% recovery (R = 0.5)	0.476	E1
B Topology	deletion fraction at 90% preservation	0.870	E2
C Control	largest actuator loss with success ≥ 0.5	0.20	E3
D Information	observed fraction for D ≤ 0.2 (ρ = 0.9)	40%	E4
E Floor	bits per coordinate at D = 0.2	1.161	E5

Table. Critical value of each layer, each taken from its own experiment.

11 Predictions, status and falsification

The table below records each hypothesis with its status after the experiments. Hypotheses that failed stay in the table, and the status column says what the evidence supports. The status is deliberately conservative.

#	Hypothesis	Status
H1	Random binary code recovery matches the closed form of Proposition 2	Supported. Max deviation 0.049 over 57 points, consistent with 400-trial sampling error
H2	Three-cluster topology survives moderate deletion	Supported to about 87% deletion
H3	Fixed-scale topology is robust to measurement jitter	Falsified. Preservation 0.51 at zero deletion with jitter 0.3
H4	Reachability requires $m \geq \lceil n/T \rceil$ actuators	Supported. Cliff at $m = 3$ as predicted
H5	Energy cost rises steeply as actuators are lost	Supported. About 1318× reference at $m = 3$, 1.0× at $m = 10$
H6	Structure lowers the observed fraction needed for estimation	Supported. 90% at $\rho = 0.2$ down to 40% at $\rho = 0.9$
H7	The composite frontier governs joint recoverability	Untested end to end. Illustrative only
H8	The Landauer-type floor constrains restoration cost	Stated as a bound. Not constraining at these scales. Not tested in a physical system

H1 and H4 are proved and checked numerically. H2, H5 and H6 are numerical results within the stated models. H3 was expected to hold and did not. H7 and H8 remain open. Falsifiability is built in: each hypothesis names a measurement that would contradict it.

12 Reproducibility

All results come from `code/recoverability_experiments.py`, which uses NumPy, SciPy and Matplotlib with one random generator seeded with 20261108. It writes JSON results to `results/` and figures to `figures/` in both PNG and SVG. `code/build_paper.py` reads those files and produces this PDF, so every number in the text is taken from the same run. Runtime on a single CPU core is a few minutes, dominated by the topology experiment. Appendix B lists every parameter.

13 Limitations

Stylised models. All systems are abstract and dimensionless. Nothing here fits biological data, and no numerical value should be read as a prediction for any organism.

Independence. The composite frontier multiplies three probabilities measured in separate simulations. A joint simulation is needed to know how far the true joint probability departs from the product.

Gaussian and linear assumptions. Layer D assumes Gaussian statistics and linear observation with known covariance. Layer C assumes linear dynamics with known matrices. Real systems are nonlinear and their parameters must be estimated.

Sample sizes. Monte Carlo estimates use 300 to 400 trials per point and 30 actuator sets, so differences smaller than a few percent are not resolved. Bands in the figures show sampling uncertainty only, not model uncertainty.

Topology. Single-linkage clustering at a fixed scale is one invariant among many. Its dependence on the cut is shown in the jitter result, and other invariants may behave differently.

Literature. The search was not exhaustive. Novelty is claimed only to our knowledge, and parts of the composition may exist elsewhere.

Connection to experiment. A biological test would need to measure information content, reachability and energy in a living system, which none of the models provides. Section 14 outlines what such a test would require.

14 Discussion

The results support a three-part picture. Coding sets how much damage a representation tolerates, and it does so with a sharp threshold that follows the rate. Structure, measured as correlation or as the topological organisation of a cloud, sets how much information a partial observation carries. Control sets whether any admissible action can reach the target, and it does so with a discrete cliff followed by a steep cost. In the models studied, the discrete reachability event sets the frontier more strongly than information loss does over the range tested. The physical floor is far below the other costs.

Three things follow for research design. First, recoverability should be measured in the three currencies together: how much information survives, how much of the control space survives, and what the restoration costs. Measuring one of them alone can mislead. Second, topological and other structural summaries must be calibrated to measurement noise before comparison, as the jitter result shows. Third, the most informative failures here were the hypothesis that jitter would not matter and the expectation that the composite would be a smooth function. Both were overturned by the numbers.

We wish to be explicit about what these results do not say. They do not show that any damaged organism, tissue or person can be restored, that identity can be preserved or recovered, or that any intervention would achieve function. They describe conditions under which an abstract information state is recoverable in a stylised model. Any statement about living systems requires a separate experiment with its own measurements, controls and ethical review. A minimal first step, outside the scope of this paper, would be to measure information retention, reachability and energy in a model organism under controlled damage, with pre-registered criteria for success and failure.

15 Conclusion

We formalised recoverability of damaged information-bearing systems as five layers, proved six propositions, and checked each layer numerically. The coding threshold, the topological signature under deletion, the reachability cliff, the dependence of estimation on structure and the thermodynamic floor are all reproducible from one script. Two findings are negative or cautionary: fixed-scale topology is fragile to noise, and the composite frontier is governed by a discrete control event. The composite conjecture remains untested end to end, and the next step is a joint simulation. The framework is a tool for asking where recoverability ends, and it is deliberately silent on whether any living system can be brought back.

Appendix A Proof details

Proof of Proposition 2. Let the k rows of G_S be $r_1, \ldots, r_k$. Each is uniform on F_2^s and independent of the others. The probability that $r_1, \ldots, r_k$ are linearly independent is found sequentially: given independence of the first i vectors, they span a subspace of size 2^i, so r_{i+1} must avoid that subspace, which has probability $1 - 2^{i-s}$. Multiplying over $i = 0, \ldots, k-1$ gives the stated product. If $s < k$ the rows cannot be independent. The limits follow because each factor tends to 1 when $s - k$ grows, and the union bound gives $P_{\text{full}} \geq 1 - 2^{k-s}$. The Monte Carlo check in Figure 2 uses independent draws of G and S, so the sampled and exact probabilities are directly comparable. □

Proof sketch of Proposition 3. The bound rank $M \leq Tm$ holds because M has Tm columns, and rank $M \leq n$ because M has n rows. For genericity, rank M is at least r exactly when some $r \times r$ minor of M is nonzero. Each minor is a polynomial in the entries of (A, B). A polynomial that is nonzero at one point is nonzero on a dense open set, so it suffices to exhibit one pair (A, B) with rank $\min(n, Tm)$. Taking A a cyclic shift and B the first $\min(m, n)$ unit vectors gives that pair, which is the standard Kalman example. The numerical cliff in Figure 4 is consistent with this, since the rank changes exactly at $m = 3$ for $n = 20$ and $T = 8$. □

Appendix B Parameter table

Experiment	Parameter	Value
E1 codes	information symbols; coded symbols; rates; erasure grid	k = 12, 24, 36; n = 48; R = 0.25, 0.5, 0.75; 0 to 0.9 step 0.05
E1 sampling	trials per point; code ensemble	400; fresh uniform random binary matrix per trial
E2 cloud	clusters; points per cluster; centres; standard deviation	3; 80; (0,0), (4,0), (2,3.5); 0.5
E2 signature	linkage; cut; minimum cluster size	single linkage; 1.0; 5
E2 damage	deletion grid; jitter; trials	0 to 0.9 step 0.05; 0 and 0.3; 300
E3 system	states n; actuators m_full; horizon T; spectral radius	20; 10; 8; 0.9
E3 sampling	actuator sets per m; targets per set; success factor κ	30; 10; 2
E4 model	states; noise variance; correlation ρ; observed fraction grid	40; 0.1; 0.2, 0.6, 0.9 (and 0 to 0.98); 0.05 to 1.0
E4 sampling	random subsets per estimate	40
E5 floor	temperature; coordinates in energy total	310 K; n = 40
Seed	single generator	20261108 (NumPy default_rng)

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